

SPATIAL EFFECTS OF DIESEL TAXES: TRADE AND EXTERNALITIES

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Abstract

Freight transported by heavy-duty trucks generates carbon, local-pollution, congestion, accident, and pavement damages along their routes. This paper quantifies the spatial externalities of trade and the welfare effects of state and federal diesel taxes. I develop a multi-commodity open-economy trade model that follows each trade flow through the U.S. highway network, allocating emissions, damages, and tax revenue to every state and county it crosses. Using Freight Analysis Framework shipment data and web-scraped diesel prices, I estimate freight responses to fuel costs and compute counterfactuals. Increasing the federal diesel tax to its 1993 real value avoids more than \$700 million in damages over 2018–2024, with pavement damage generating the largest benefit. Most benefits occur in corridor states that freight passes through, and part of the welfare cost is borne by foreign buyers of U.S. exports. A unilateral state diesel tax creates large spillovers: when New Mexico raises its tax to California’s level, national pollution falls, but freight shifts to alternative corridors, increasing externalities in neighboring states. These findings highlight the importance of coordinating state and federal diesel tax policies.

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1 Introduction

Freight shipment via heavy-duty trucks generates carbon, local-pollution, congestion, accident, and pavement costs along the routes it travels. This paper builds a framework for quantifying the externalities of trade and analyzing the welfare effects of diesel taxes. I build a multi-commodity, open-economy general-equilibrium model of trade on the U.S. road network, combine data on freight flows and route-level diesel prices, and estimate the fuel-cost elasticity of trade separately for each commodity from gravity regressions. The model tracks each shipment along the highway route it actually travels, so emissions, damages, and tax revenue are assigned to every state and county the freight crosses, not only its origin and destination. I use the model to study two counterfactuals: restoring the federal diesel tax to its 1993 real value, and a unilateral increase in a single state's diesel tax.

The first counterfactual asks how welfare and the environment would change if the federal diesel tax, fixed in nominal terms since 1993, were returned to its historical real value over 2018–2024. The policy avoids more than \$700 million in external damages. Most of this is not carbon: congestion, accident, and pavement wear together account for \$314 million, more than the \$298 million in avoided climate damage valued at a \$190 social cost of carbon, and local air pollution adds another \$104 million. Adding these non-carbon externalities more than doubles the measured benefit of the tax.

The framework also shows how these gains are split across states. The benefits fall mostly on the corridor states that freight only crosses: a state such as Louisiana avoids a large volume of emissions and collects substantial revenue on trucks that neither start nor end there, while the real-income cost falls on the states that consume the goods. Because U.S. export varieties are differentiated, part of the tax is paid by foreign buyers on the domestic leg of export shipments, so the revenue the tax raises is larger than the burden on U.S. residents. Looking within states changes the picture too: in the densest states, such as California, local-pollution damage rises even as it falls nationally, because the damage from a pollutant depends on where it is emitted.

The second counterfactual shows that a state acting on its own creates spillovers onto other states. When New Mexico raises its diesel tax to California’s level, it gains: the higher cost of crossing the state cuts the through-freight that pollutes its corridors, and it still collects revenue on the freight that keeps crossing. National pollution falls too, because freight demand responds to the higher shipping cost. But the drop is not shared evenly. As buyers switch to suppliers whose freight travels other routes, a set of Plains and Midwest states takes on more freight and sees its own pollution rise. A state that raises its tax makes itself better off and its neighbors worse off.

The model builds on (Armington, 1969). It is an open economy that trades through its ports and land border crossings: imports enter and exports leave at these gateways and then travel the domestic network, so international shipments, like domestic ones, generate ton-miles and externalities on U.S. soil. I solve the counterfactuals in changes using exact hat algebra; similar assumptions would give similar welfare numbers under other leading trade theories (Arkolakis, Costinot, and Rodríguez-Clare, 2012).

To estimate the model’s key parameter, I run gravity regressions that recover the fuel-cost elasticity of trade separately for each of seven commodity groups. This elasticity measures how strongly freight in each commodity responds to the diesel price, and it is identified from within-route variation in fuel costs, after absorbing origin, destination, and route fixed effects. The estimates are largest for low-value, fuel-intensive bulk goods such as construction materials and food, and small or near zero for high-value manufactures. This fits the idea that fuel is a bigger share of the delivered price for heavy, cheap goods.

Taking the model to the data means locating every shipment on the road. I combine Freight Analysis Framework shipment data (Freight Analysis Framework, 2024) with web-scraped, route-level diesel prices and a graph of the U.S. highway system that I build so that each origin–destination flow can be traced through the states and counties it crosses. To my knowledge, this is the first trade model of U.S. freight that follows shipments to the county level along their routes. That detail is what lets me assign emissions and damages to transit

states and value local pollution with county-specific damages.

This paper contributes to three literatures. It is closest to the trade-and-environment literature that adds emissions to a structural gravity model, in particular Shapiro (2016), and uses the same tools (Eaton and Kortum, 2002; Alvarez and Lucas, 2007; Head and Mayer, 2014). Unlike most of that literature, which looks at the effect of trade on productivity or of environmental regulation on growth (Copeland and Taylor, 2004; Frankel and Rose, 2005; Cherniwchan, Copeland, and Taylor, 2017), I focus on the externalities that trucking generates within the United States. The difference is spatial: because I place each shipment on its route, I assign emissions, damages, and revenue to every state and county a truck crosses, which lets me measure the externality a transit state experiences from freight it neither produces nor consumes.

It also builds on the literature on trucking externalities (Leard et al., 2015; Cohen and Roth, 2016; Muehlenbachs, Staubli, and Chu, 2017), and most directly on Parry (2008), who computes an optimal diesel tax (\$1.12 per gallon) that internalizes congestion, accidents, pavement damage, noise, energy security, and pollution. I extend this work in two ways. First, I put these externalities inside a general-equilibrium trade model, so the tax changes not just the level of trucking but the trade flows and real income; since transport costs are a large share of trade costs, up to 30 percent (World Trade Organization, 2019), these effects matter. Second, the spatial structure shows that the congestion, accident, and pavement externalities fall along the corridors the freight crosses, not where the goods are bought.

Finally, the paper connects to work on the local pollution of transportation (Small and Kazimi, 1995; Brodeur, Cook, and Wright, 2021; Rivera, 2021). Closest is Holland et al. (2016), who estimate the county-level environmental benefits of electric vehicles; I use a similar integrated-assessment approach, so the local benefit varies across counties with population density and baseline pollution. In terms of method, the model sits between two approaches: reduced-form studies that leave out general-equilibrium forces, and large computable general-equilibrium models. It uses a handful of equations and one estimated elas-

ticity per commodity, but still produces welfare and environmental results at a fine spatial scale.

The paper proceeds as follows. Section 2 describes the institutional background. Section 3 describes the data and the construction of the road network. Section 4 presents the model. Section 5 sets out the counterfactual method. Section 6 reports the fuel-cost elasticity estimates. Section 7 presents the federal and New Mexico counterfactuals. Section 8 concludes.

2 Institutional background

Trucks are an important part the U.S. freight system: they move 72.2 percent of domestic freight by weight and 80.8 percent by value, and the U.S. Department of Transportation projects a further 50 percent increase in truck ton-miles by 2050 (American Trucking Associations, 2023). This freight is not evenly distributed across states and routes. More than 72 million Americans live close to trucking corridors, and they are disproportionately lower-income and members of communities of color (Tessum et al., 2021; Demetillo et al., 2021). Heavy-duty trucks account for 32 percent of on-road NO_x emissions (a principal precursor of $PM_{2.5}$) and 23 percent of on-road greenhouse-gas emissions (Clean Trucks Plan, 2021).

In August 2021, the EPA set new NO_x and criteria-pollutant standards for medium- and heavy-duty vehicles beginning with model year 2027, and in April 2023 the White House announced a rule targeting 25 % all-electric heavy-truck sales by 2032. The changes in policies happened not only on the federal level: 11 states raised their diesel taxes in 2023. A state diesel tax is similar to a gasoline tax in theory, but in practice its welfare consequences are more complex, because trucking is mostly interstate. Although 90% of household car trips in 2017 were 15 miles or less (National Household Travel Survey, 2017), the average heavy-truck shipment traveled 857 miles, and interstate shipments made up 56% of the value of goods moved by truck. A state that raises its diesel tax, therefore, affects freight that

mostly neither originates nor terminates within it.

In this paper, I investigate the effects of change in fuel costs which are the second highest costs for carriers after driver compensation and comprise almost a quarter of the total average marginal costs per mile (Williams and Murray (2020)). The trucking industry annually spends almost \$105 billion on diesel fuel while the average trucker spends about \$70,000. However, only owner-operators pay these costs by themselves. The rest of the truckers (more than 60 %), company drivers, usually have their fuel expenses covered by their employer either directly with a fuel card issued by the employer or with the reimbursement in a paycheck.

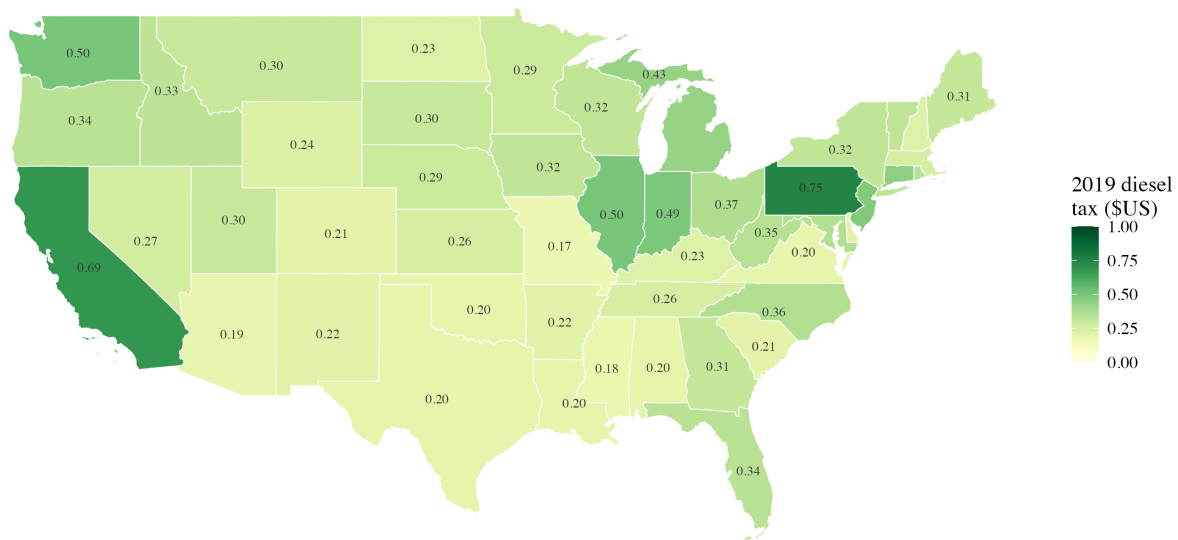


Figure 1: Total diesel state taxes in 2019 (*Source: EIA*)

Up to 30% of the fuel costs are determined by diesel taxes. In the US diesel tax is collected on the federal and state level. The federal tax has remained at \$0.244 per gallon since 1993. The state taxes, however, vary every year and substantially differ across states. The International Fuel Tax Agreement (IFTA) requires operators to pay the state diesel tax

even if they just drove right through and did not buy any fuel there: drivers have to record mileage driven in each state and pay a state diesel tax equivalent to the portion of the fuel they used while driving through the state. In addition to the state tax, states may charge sales and/or use taxes, inspection fees, environmental fees, or other fees. I define the total state diesel tax as the sum of the state tax and the rest of the fees. Figure 1 illustrates the difference in total state taxes across the US: in 2019 the lowest total state tax was charged by Missouri (\$0.17) and the highest by Pennsylvania (\$0.75).

This paper assumes that the increase in diesel tax affects real income only through changes in trade. One might come up with two main critiques of this assumption. The first one is that the diesel tax can affect the agriculture or construction sectors since diesel is the main source of fuel for the equipment and vehicles involved in production. However, the diesel fuel which is sold for "off-road" purposes is not subject to state and federal excise taxes. To distinguish the off-road diesel from the on-road (the two are chemically identical), the first one is dyed red. The fine for illegal use of dyed diesel on highways is \$10/gallon or \$1,000 per violation.

The second margin that this paper does not account for is a dead weight loss for diesel-powered car owners created by the increase in diesel tax. While this margin is substantial in the context of the EU where diesel-powered vehicles constitute 50% of the total sales, it is not a case in the US since this type of vehicles constitutes only 3% of the car fleet (Chambers and Schmitt (2015)).

3 Data and Graph description

3.1 Trade flows

I use data from Freight Analysis Framework (2024) released by the Bureau of Transportation Statistics, Office of the Assistant Secretary for Research and Technology and Federal Highway Administration U.S. This dataset is similar to the Commodity Flow Survey (CFS) that is

commonly used in trade literature. However, the CFS primarily covers only the mining, manufacturing, and wholesale sectors. In contrast, FAF is nationally representative and integrates additional data to estimate volumes of shipments from industries that are not covered by the CFS, including import and export. Table 1 describes the data the FAF provides for each origin-destination pair.

Table 1: Freight Analysis Framework (FAF5.5) data description

Data	Description
Domestic origin	FAF region where a freight movement begins (the domestic portion of) the trip
Domestic destination	FAF region where a freight movement ends (the domestic portion of) the trip
Foreign origin	For imports, the foreign region where a freight movement begins
Foreign destination	For exports, the foreign region where a freight movement ends
Mode of transportation	One of the seven modes of transportation
Commodity type	2-digit level of the Standard Classification of Transported Goods (SCTG)
Tons	Total weight of commodities shipped
Tons-miles	Total weight of commodities shipped multiplied by miles traveled
Value	Total value (in 2017 dollars) of commodities shipped
Current value	Total value (in current dollars) of commodities shipped

FAF reports each origin–destination flow by 2-digit SCTG commodity and by year. I use the annual data from 2018 to 2024 and keep the commodity dimension throughout: both the gravity estimation and the counterfactuals are run at the commodity level, with goods grouped into end-use categories (Section 6.3). This allows the response of freight to the diesel price differ across commodities, from fuel-intensive bulk goods to high-value manufactures.

FAF5.5 provides data for the 132 domestic regions. The domestic regions are divided into the state portions of large metropolitan areas and remainders of states. Metropolitan areas consist of Metropolitan Statistical Areas or Consolidated Statistical Areas as defined by the Office of Management and Budget. When a metropolitan area lies entirely within a state or when a state’s portion of a multi-state metropolitan area is large enough to support the sampling procedures in the CFS, the area becomes a separate FAF region. Small single-state metropolitan areas and small portions of a multi-state metropolitan area are part of the State or the Remainder of the State.

I construct a graph that is based on the US highway system and represents the movement

of the trade flows across US states¹. The goal of this exercise is to enrich the data on origin-destination pairs of trade flow. First, FAF5.5 does not provide data on the exact route of the freight movement which is essential for the externalities analysis. Second, one needs to know fuel costs along the route to be able to construct the proxies for the trade costs.

In this paper, I use 3 terms to refer to the parts of the graph: node, leg, and route. A *node* is an origin/destination region. A *leg* is a graph segment that connects two nodes. A *route* is a combination of legs. Figure 2 provides an example of the route. The route from Phoenix, AZ to Albany, NY consists of 13 legs and goes through 14 nodes and 10 states.

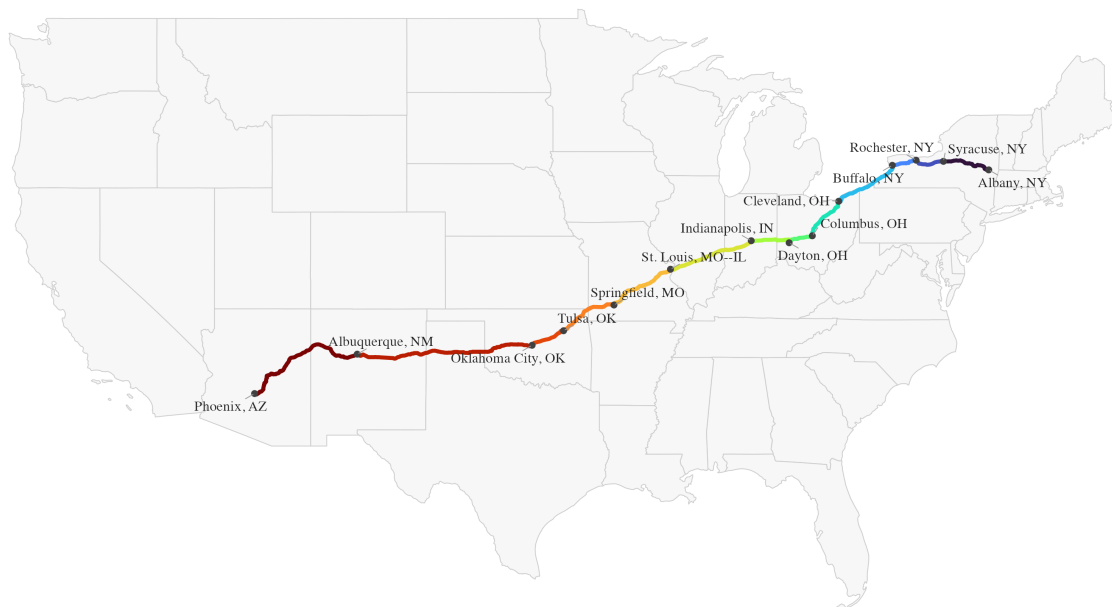


Figure 2: Route from Phoenix to Albany

Some of the FAF regions are modified because of two reasons. First, several FAF regions represent multi-state metropolitan areas (such as "Portland-Vancouver-Salem, OR-WA CFS Area (OR Part)" and "Portland-Vancouver-Salem, OR-WA CFS Area (WA Part)") which

¹I explicitly exclude 2 states from the analysis: Hawaii and Alaska. Rhode Island and New Jersey are not separately represented in the dataset and their trade volumes are merged with data on New York, Massachusetts, and Pennsylvania

complicates the state-level counterfactual analysis. Second, the "State" (for example, "Vermont") or "Remainder of State" (for example, "Remainder of Washington") regions do not allow one to geolocate the origin/destination.

I use the following algorithm to modify these regions: first, I overlay the shapefiles of FAF areas and the US urban areas created by the U.S. Census Bureau, Department of Commerce and choose the urban area with the largest population for each CFS area. That allows me to geolocate the first set of "problematic" FAF regions. Second, for "State" or "Remainder of the State" FAF regions I choose an urban area in the state with the biggest population and that was not included in the graph before: for example, Burlington, VT for Vermont and Kennewick–Richland–Pasco, WA for Remainder of Washington (since Seattle, WA was included before). These nodes account for the rest of the state.

To assign a route to an origin-destination pair I use the fastest route offered by Google Maps. I define the length of the leg as the distance between the borders of the nodes. Figure 3 shows the completed graph. It consists of 114 nodes (convex hulls of the urban areas' shapefiles) and 483 legs (dark lines) that generate $\frac{114*113}{2} = 6441$ routes. The longest route (the route from San Diego, California to Portland, Maine) consists of 16 legs.

I also create adjustment factors that represent the average distance the truck spends in the area when it goes through it. The length of the route is equal to the sum of legs and adjustment factors that connect the legs plus half of the sum of origin-destination adjustment factors.

FAF also records the imports and exports that move by truck, with the foreign region of origin or destination. I attach each international shipment to the gateway (the port or land border crossing) through which it enters or leaves the country, and route it over the domestic network from that gateway. The gateways are therefore regular nodes of the graph that additionally carry the US legs of trade flows, which is what makes the network an open economy in Section 4.

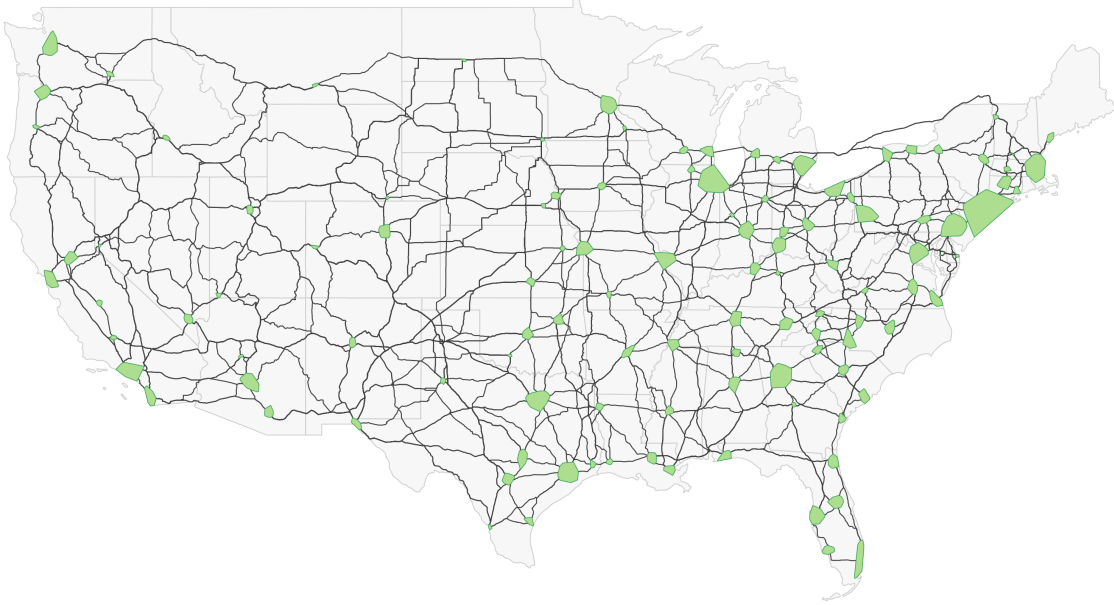


Figure 3: Graph of trade flows and urban areas

Routing I choose the fastest route for shipment. Heavy trucks concentrate on the Interstate system, both because they are restricted or discouraged on many arterial roads and because the return on a carrier’s two most expensive inputs (the driver and the tractor-trailer) depends on how many loads it completes. Fuel, by contrast, is a minor cost: it averaged \$0.48 per mile in 2024, about 21 percent of the \$2.26 average marginal cost, always well below driver wages and benefits of roughly \$1.00 per mile (Leslie and Murray (2025)).² Because fuel is a small share of the cost of a mile and the choice of corridor is governed by time and network restrictions rather than fuel, the fastest route returned by Google Maps is a close proxy for the corridor a truck actually travels.

²Fuel ranged from 18.7 percent of marginal cost in 2020 to 28.5 percent in 2022; driver wages and benefits were 40–45 percent of marginal cost throughout (Leslie and Murray (2025), Table 8).

3.2 Diesel prices

Diesel price data are web-scraped from daily Gas Price Charts (2026) published by Gas Buddy. Given the Gas Buddy (GB) data are self-reported, the data accuracy may vary across gas stations. Table 2 compares the diesel prices from GasBuddy with the diesel prices published by the Energy Information Administration for 7 Petroleum Administration for Defense Districts (PADD) regions.

Table 2: Comparison of Gas Buddy prices with EIA (2019)

Source	New Engl.	Centr. Atlant.	Low. Atlant.	Midwest	Gulf Coast	Rocky Mount.	West Coast
EIA	2.723	2.806	2.48	2.426	2.313	2.517	3.084
GasBuddy	2.696	2.779	2.478	2.438	2.241	2.436	3.194

Notes: State-level prices from Gas Buddy are weighted by population to match Petroleum Administration for Defense Districts of EIA. EIA shows volume-weighted average diesel price

I geolocate GB stations and assign its data to the urban areas. Figure 4 shows the geographic distribution of GB gas stations. 451 GB stations include data on 379 urban areas (for example, Spokane, WA), 25 cities (for example, Vancouver, WA), 34 counties (for example, Asotin County, WA), and 13 duplicate areas (for example, gas stations in Seattle, WA and Tacoma, WA are located in one urban area, Seattle-Tacoma, WA).³ For the duplicate areas, I use the data from the gas stations that are located in a more populated area (for example, in the case of Seattle-Tacoma, WA it's Seattle, WA). I use shapefiles for counties and cities from the US census to geolocate the GB gas stations that are not located in urban areas. For 5 urban areas in the graph that do not have GB stations assigned to them, I assigned state-level diesel prices. In total, I use the data from 443 GB locations.

Next, I construct the distance-weighted price for every origin-destination pair:

$$\bar{P} = \frac{\sum_{i=1}^{N-1} (D_{i+1} - D_i)(P_{i+1} + P_i)}{2 * TotalDistance} \quad (1)$$

For example, trade from San Diego to El Paso goes through two urban areas with GB

³The 451 stations resolve to 443 distinct geolocated locations after merging duplicates; the map in Figure 4 shows 442, dropping one Texarkana, TX shape that coincides with Texarkana, AR.

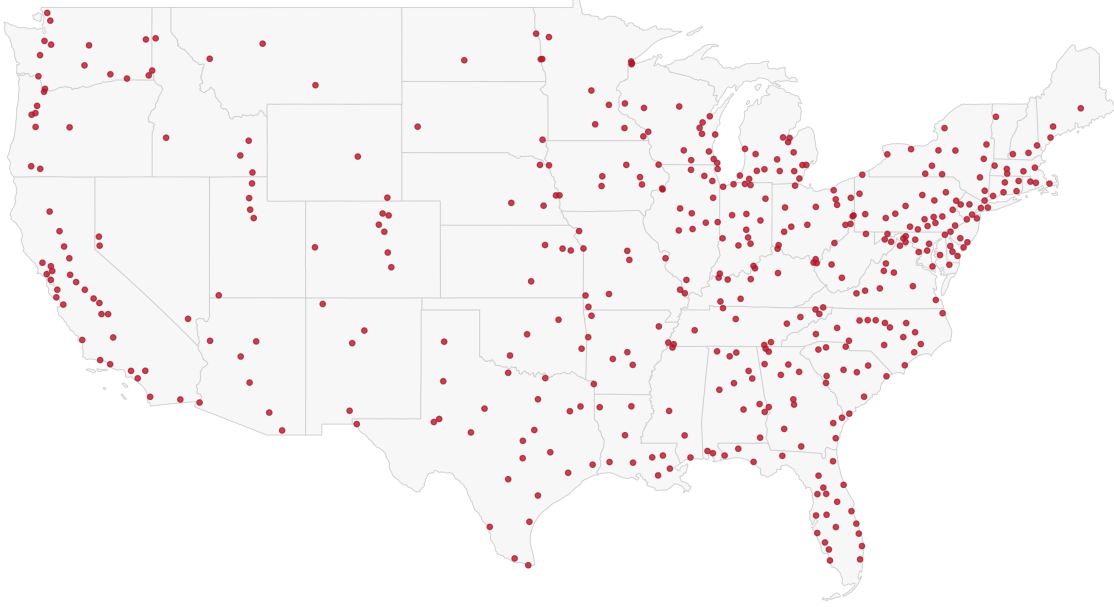


Figure 4: Geographic distribution of GasBuddy stations (Source: Gas Buddy)

stations: Yuma and Tucson. Then the route-average price is:

$$\begin{aligned} \bar{P} = & P_{SanDiego} \frac{Yuma - SanDiego}{2TotalDistance} + P_{Yuma} \left(\frac{Yuma - SanDiego}{2TotalDistance} + \frac{Tucson - Yuma}{2TotalDistance} \right) + \\ & + P_{Tucson} \left(\frac{Tucson - Yuma}{2TotalDistance} + \frac{ElPaso - Tucson}{2TotalDistance} \right) + P_{ElPaso} \frac{ElPaso - Tucson}{2TotalDistance} \end{aligned} \quad (2)$$

4 A Multi-Commodity Open-Economy Model of Trade and the Environment

I use a multi-sector Armington model combined with the US road network. I add the US road network to the standard Armington model in order to assign every trade flow to the states and counties it crosses. The model has three important features. First, the trade

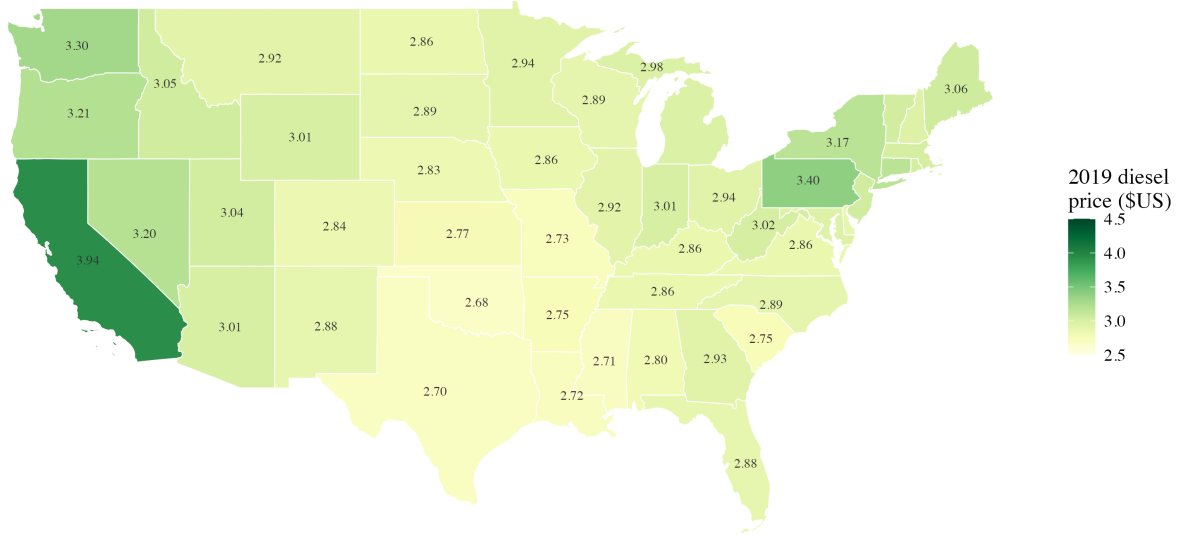


Figure 5: Distribution of diesel prices in 2019 (Source: Gas Buddy)

flows are disaggregated into K commodities, each with its own elasticity of substitution and its own trade-cost elasticity, so that the response of freight to the diesel price is allowed to differ across goods. Second, the economy is open through its ports and land border crossings: imported varieties are available at these gateways at fixed prices, and foreign buyers demand U.S. goods delivered to the same gateways. Imports and exports, like domestic shipments, generate ton-miles on the routes they travel on U.S. soil, and therefore generate externalities there. Third, the tax revenue and the pollution appear in every state a shipment passes through, not only to its origin and destination. This is what allows the model to capture the externalities a state experiences from freight that it neither sends nor receives.

4.1 Environment

There are N domestic areas indexed by $i, j \in \mathcal{D}$, the urban areas that correspond to the nodes of the graph. A subset $\mathcal{G} \subseteq \mathcal{D}$ of these areas are *gateways*: the ports and land border crossings through which the country trades with the rest of the world. The rest of the world enters the model only through these gateways. For each commodity an imported variety is available at a gateway at an exogenous price, and each gateway carries an exogenous foreign demand for U.S. goods. Goods are grouped into K commodities indexed by k , following the SCTG classification. Labor is the only production factor and is immobile across areas. Area i has labor supply L_i and wage w_i , and a worker in area i produces A_i^k units of commodity k . Time is indexed by $t \in \{2018, \dots, 2024\}$; I suppress the year subscript where it is not essential.

4.2 Preferences and the commodity-level gravity equation

Consumers in area j have two-tier preferences. The upper tier is Cobb–Douglas across commodities, with expenditure shares α_j^k that sum to one. The lower tier is CES within each commodity over the origins of that good, with an elasticity of substitution $\sigma_k > 1$ that is specific to the commodity. Within commodity k , a buyer in j chooses among domestic producers $i \in \mathcal{D}$ and imported varieties, which I index by their entry gateway. Let $\mathcal{M} = \{m_g : g \in \mathcal{G}\}$ collect the import sources and let $\mathcal{O} = \mathcal{D} \cup \mathcal{M}$ be the full set of origins facing a domestic buyer. Preferences are

$$U_j = \prod_{k=1}^K (C_j^k)^{\alpha_j^k}, \quad C_j^k = \left[\sum_{o \in \mathcal{O}} (c_{oj}^k)^{\frac{\sigma_k - 1}{\sigma_k}} \right]^{\frac{\sigma_k}{\sigma_k - 1}}. \quad (3)$$

Cobb–Douglas separability assigns a constant share of expenditure to each commodity, $E_j^k = \alpha_j^k E_j$, and within a commodity the problem is the standard CES allocation. Minimizing

expenditure (I provide the steps in Appendix B) gives the commodity-level gravity equation

$$X_{oj}^k = \left(\frac{\tau_{oj}^k p_o^k}{P_j^k} \right)^{1-\sigma_k} E_j^k, \quad P_j^k = \left[\sum_{o \in \mathcal{O}} (\tau_{oj}^k p_o^k)^{1-\sigma_k} \right]^{\frac{1}{1-\sigma_k}}, \quad (4)$$

where $X_{oj}^k = p_{oj}^k c_{oj}^k$ is the value of area j 's purchases of commodity k from origin o , $p_{oj}^k = \tau_{oj}^k p_o^k$ is the delivered price, and P_j^k is the price index for commodity k . The origin price p_o^k is an endogenous price when o is a domestic producer and a fixed price when o is an import source, as I specify below.

4.3 Firms and trade costs

Each domestic commodity market is competitive, so the price equals the marginal cost, $p_i^k = w_i/A_i^k$ for $i \in \mathcal{D}$. An import source carries a fixed price, $p_{m_g}^k = \bar{p}_g^k$, which equals the world price plus the cost of the international leg into gateway g : the foreign-soil transport, the maritime or air freight, the port handling, and any tariff. This price is exogenous to U.S. diesel policy. I model the iceberg trade cost so that it depends on the route distance and the route diesel price, with commodity-specific elasticities,

$$\tau_{oj,t}^k = \text{Dist}_{oj}^{\beta_d^k} \rho_{oj,t}^{\beta_p^k}, \quad (5)$$

where $\rho_{oj,t}$ is the route diesel price, that is, the mileage-weighted average of the per-gallon diesel prices along the route. The route diesel price ρ is distinct from the goods prices p_o^k . For a domestic origin the leg is the full route $o \rightarrow j$, and for an import source it is the gateway-to-destination route $g \rightarrow j$. Because I denote the international leg as the fixed price $\bar{p}_g^k$, the diesel tax touches only the U.S. leg, and I never need the level of the international leg. The fuel-cost elasticity β_p^k is the parameter of interest. Intuitively, it is the share of fuel in the delivered price, so low-value, fuel-intensive bulk commodities respond more strongly to the diesel price than high-value goods do. I recover the coefficients from the within-route gravity in Section 6. I apply the same fuel-cost elasticity β_p^k to all three U.S. legs (domestic, import,

and export) since each is U.S.-soil trucking of commodity k ; the coefficient is estimated from domestic freight and taken to govern the import and export legs as well.

4.4 Open economy: imports and exports at gateways

Imports The gateways are U.S. nodes. Each gateway g supplies, for each commodity, an imported variety at the fixed price $\bar{p}_g^k$. The variety competes in a domestic buyer's CES nest through its delivered price $\bar{p}_g^k \tau_{gj}^k$, and only the domestic leg τ_{gj}^k responds to the diesel price. The shipment generates ton-miles in every state on that leg:

$$\text{an import to } j \text{ through } g \text{ crosses state } s \iff s \in \text{route}_{g \rightarrow j}. \quad (6)$$

Exports Each gateway g carries an exogenous foreign demand $E_g^{k,X}$ for U.S. goods. Foreign buyers split this demand, with elasticity σ_k , between U.S. varieties delivered to g and a fixed-price outside option $\bar{p}^{k*}$, so U.S. producers face a downward-sloping export demand,

$$X_{ig}^{k,X} = \left(\frac{\tau_{ig}^k p_i^k}{P_g^{k,X}} \right)^{1-\sigma_k} E_g^{k,X}, \quad P_g^{k,X} = \left[\sum_{i \in \mathcal{D}} (\tau_{ig}^k p_i^k)^{1-\sigma_k} + (\bar{p}^{k*})^{1-\sigma_k} \right]^{\frac{1}{1-\sigma_k}}. \quad (7)$$

Producer i pays the domestic leg τ_{ig}^k to reach the gateway, and the shipment crosses the states on route $i \rightarrow g$. The outside option is what lets U.S. export volume fall when the diesel tax raises the cost of reaching the gateway.

Through-traffic A truck burns diesel in every state along its route, not only in its origin and destination states. A state therefore experiences pollution and collects tax revenue from freight that only passes through it. This through-traffic is the main mechanism behind the state-level results: a landlocked state such as New Mexico experiences the pollution of goods that are shipped between other states. The model includes domestic, import, and export shipments (FAF trade types 1, 2, and 3); it does not include foreign-to-foreign transit.

Closure The rest of the world is exogenous to U.S. diesel policy. I hold the import prices and the foreign export demand fixed in the counterfactuals, so that $\hat{p}_g^k = 1$ and $\hat{E}_g^{k,X} = 1$. The equilibrium system is then defined over the domestic areas alone, with no foreign market to clear, and the rest of the world is summarized entirely by the fixed gateway prices and demands. I carry the baseline U.S. trade imbalance as a fixed deficit, so that the observed flows form a consistent baseline equilibrium.

4.5 Tax revenue

Every state collects a diesel tax on the fuel burned within its borders. Under the International Fuel Tax Agreement a carrier pays the state tax in proportion to the miles it drives in each state, so a state collects revenue on every truck that crosses it, including freight that neither originates nor terminates there. The relevant tonnage is that of every U.S. truck leg: domestic, import, and export. Writing c_{rs}^k for the tonnage of commodity k routed through state s on truck leg r , the gallons burned in state s are

$$G_s = \sum_{k=1}^K \sum_r \frac{c_{rs}^k \text{miles}_{rs}}{e_k}, \quad (8)$$

where miles_{rs} is leg r 's in-state mileage in s and e_k is the freight efficiency in ton-miles per gallon. When commodity-specific efficiencies are not available I use a common value of $e = 60$ ton-miles per gallon. The federal tax is collected on the same gallons. I attribute revenue to the state where the fuel is burned, at both the state and the federal rate,

$$T_s = \text{tax}_s G_s, \quad T_s^{\text{fed}} = \text{tax}_{\text{fed}} G_s, \quad T_{\text{fed}} = \sum_s T_s^{\text{fed}} = \text{tax}_{\text{fed}} \sum_s G_s, \quad (9)$$

so that a corridor state's revenue depends on the freight that crosses it rather than on the goods its residents buy.

4.6 Pollution and other externalities

The emissions are linear in ton-miles and I count them on every U.S. truck leg. The combustion of diesel produces a fixed amount of CO_2 per gallon, so the carbon is tied to the same efficiency e_k that governs the gallons in (8). With $\psi = 10,180$ grams of CO_2 per gallon (EPA),

$$CO_2 = \psi \sum_k \sum_r \sum_s \frac{c_{rs}^k \text{miles}_{rs}}{e_k}, \quad TSCC = SCC \cdot CO_2, \quad TSCC_s = \omega_s TSCC. \quad (10)$$

The implied factor is ψ/e_k grams per ton-mile. With the common efficiency $e = 60$ this equals 170 grams of CO_2 per ton-mile, the EPA value for combination trucks reported in Table 3.⁴ I value the carbon at the EPA social cost of carbon of \$190 per ton.

I quantify the volume of local pollutants ($PM_{2.5}$, NO_x , and VOC) with the MOVES3 emission factors per ton-mile in Table 3. I aggregate them to the county and map them to monetary damages with the AP4 integrated-assessment model, updated to 2019 population and mortality. AP4 reports a marginal damage for each pollutant emitted in each county, so the local-pollution damage is resolved at the county level,

$$\text{LocalPollution}_c = \sum_k \sum_r \left(\sum_p \text{EF}_p^k d_c^p \right) c_{rc}^k \text{miles}_{rc}, \quad (11)$$

where d_c^p is the AP4 marginal damage of pollutant p emitted in county c , c_{rc}^k is the tonnage of commodity k routed through county c on leg r , and miles_{rc} is its in-county mileage. I aggregate the county damages to the state, $\text{LocalPollution}_s = \sum_{c \in s} \text{LocalPollution}_c$, for the welfare accounting and the by-state tables. The carbon term in (10) is spatially uniform; it is the county-level local term that makes a through-traffic county (and the state that contains it) experience damage from freight it neither sends nor receives.

⁴I compute the gallons that determine the tax revenue and the CO_2 that determines the carbon damages from the same efficiency, which keeps the two consistent: a truck that emits 170 grams of CO_2 per ton-mile burns $170/10,180 \approx 1/60$ of a gallon. The value $e = 60$ reflects an average load with empty return trips rather than a fully loaded truck.

Trucking also creates costs that do not depend on the fuel: traffic congestion, accidents, and pavement damage. These costs depend on the ton-miles traveled. I value them with the per-ton-mile estimates from Austin (2015), which place congestion, accidents, and pavement damage together at about 3.1 cents per ton-mile:

$$\text{NonPollution}_s = \sum_k \sum_r (\kappa^{\text{cong}} + \kappa^{\text{acc}} + \kappa^{\text{pav}}) c_{rs}^k \text{miles}_{rs}, \quad (12)$$

where κ^{cong} , κ^{acc} , and κ^{pav} are the per-ton-mile costs of congestion, accidents, and pavement damage. Like the local pollutants, these costs accumulate to the states a shipment crosses.

Table 3: Average emission factors

Pollutant	CO_2	$PM_{2.5}$	NO_x	VOC
Emission factor (g/ton-mile)	170	0.007	0.38	0.015

Notes: Emission factors in grams per ton-mile. The CO_2 factor of 170 is the EPA value for combination trucks and corresponds to the efficiency $e = 60$ ton-miles per gallon used in (8) and (10). The local-pollutant factors are MOVES3 averages, converted from grams per mile to grams per ton-mile by dividing by the average FAF load of 17.53 tons.

4.7 Welfare

Domestic welfare The real income in area j is $I_j = Y_j/P_j$, where $P_j = \prod_k (P_j^k)^{\alpha_j^k}$ is the Cobb–Douglas aggregate price index. The welfare of a state is its real income plus tax revenue net of the pollution and the non-pollution damages:

$$W_s = I_s + T_s - \text{TSCC}_s - \text{LocalPollution}_s - \text{NonPollution}_s. \quad (13)$$

Foreign welfare Foreign buyers at gateway g spend a fixed budget $E_g^{k,X}$ and, by (7), obtain real consumption $E_g^{k,X}/P_g^{k,X}$; with the budget fixed ($\hat{E}_g^{k,X} = 1$), their welfare change is the change in the gateway price index alone. The diesel tax raises the route fuel price by $\hat{\rho}_{ig}^k$, which passes into the delivered price of U.S. variety i through the fuel-cost elasticity β_p^k

of (5), so $\hat{\tau}_{ig}^k = (\hat{\rho}_{ig}^k)^{\beta_p^k}$. Writing the gateway index of (7) in changes,

$$\hat{W}_g^{k,X} = \frac{1}{\hat{P}_g^{k,X}}, \quad (\hat{P}_g^{k,X})^{1-\sigma_k} = s_g^k \sum_{i \in \mathcal{D}} \omega_{ig}^k [(\hat{\rho}_{ig}^k)^{\beta_p^k} \hat{p}_i^k]^{1-\sigma_k} + (1 - s_g^k), \quad (14)$$

where $s_g^k = \sum_i (\tau_{ig}^k p_i^k)^{1-\sigma_k} / (P_g^{k,X})^{1-\sigma_k}$ is the U.S. share of gateway g 's absorption, $1 - s_g^k = (\bar{p}^{k*})^{1-\sigma_k} / (P_g^{k,X})^{1-\sigma_k}$ is the outside-option share, and ω_{ig}^k is variety i 's share among the U.S. varieties in the nest. Foreign welfare falls with the fuel-cost pass-through: a larger β_p^k makes the delivered price of U.S. goods more sensitive to the diesel tax, raises $\hat{P}_g^{k,X}$, and deepens the rest-of-world loss. Aggregating,

$$\Delta W^* = \sum_{g,k} E_g^{k,X} \left(\frac{1}{\hat{P}_g^{k,X}} - 1 \right). \quad (15)$$

The foreign welfare change in (15) is the price-channel effect: foreign buyers face higher delivered prices for U.S. goods. I do not value the environmental component of foreign welfare. The model takes into account the domestic leg of each export up to the gateway but not its onward movement abroad, whose routing and mode are unobserved, so changes in foreign pollution lie outside the model. The environmental accounting throughout is therefore focused on emissions on U.S. territory.

On the import side the world price $\bar{p}_g^k$ is fixed, so the diesel tax raises only the domestic delivery leg and is borne entirely by U.S. buyers; foreign sellers, who receive the fixed gateway price, are unaffected. The foreign-welfare channel therefore operates only through exports.

5 Counterfactual Analysis

I solve the counterfactuals in changes, using exact hat algebra. Let $\lambda_{oj}^k = X_{oj}^k / E_j^k$ be a baseline expenditure share and $\hat{\tau}_{oj}^k = \tau_{oj}^{k'}/\tau_{oj}^k$ a trade-cost shock. The domestic prices change

to $\hat{p}_i^k = \hat{w}_i$, and the external prices are fixed ($\hat{p}_g^k = 1$), so the new expenditure share is

$$\lambda_{oj}^{k'} = \frac{\lambda_{oj}^k (\hat{\tau}_{oj}^k \hat{p}_o^k)^{1-\sigma_k}}{\sum_{o' \in \mathcal{O}} \lambda_{o'j}^k (\hat{\tau}_{o'j}^k \hat{p}_{o'}^k)^{1-\sigma_k}}, \quad \hat{p}_{m_g}^k = 1 \quad \forall g \in \mathcal{G}, \quad (16)$$

and the export shares update in the same way through (7), with $\hat{p}^{k*} = 1$ and $\hat{E}_g^{k,X} = 1$. The domestic labor-market clearing condition sums over commodities, over domestic destinations, and over the foreign demand at the gateways,

$$\hat{Y}_i Y_i = \sum_k \left(\sum_{j \in \mathcal{D}} \lambda_{ij}^{k'} E_j^{k'} + \sum_{g \in \mathcal{G}} X_{ig}^{k,X'} \right). \quad (17)$$

With the external prices fixed and the baseline deficit held constant, this determines the income changes $\{\hat{Y}_i\}_{i \in \mathcal{D}}$ up to a numeraire. The diesel-tax counterfactual enters only through the price channel: the new route price adds the tax change to the fuel cost on each U.S. leg,

$$\hat{\tau}_{oj}^k = \left(\frac{\rho'_{oj}}{\rho_{oj}} \right)^{\beta_p^k}, \quad (18)$$

applied to the full route $o \rightarrow j$ for domestic shipments, the gateway leg $g \rightarrow j$ for imports, and $i \rightarrow g$ for exports, while the international leg is unchanged.

Real income The multi-sector exact-hat result gives the change in real income as the expenditure-weighted product of the changes in the own shares, each carried at its own elasticity,

$$\hat{I}_j = \prod_k (\hat{\lambda}_{jj}^k)^{\frac{\alpha_j^k}{1-\sigma_k}}. \quad (19)$$

Each commodity enters with its own exponent $1/(1-\sigma_k)$, where $(1-\sigma_k)$ is the trade elasticity of commodity k and $1/(1-\sigma_k)$ is the welfare exponent. Equation (19) is the sufficient statistic when income equals labor income; with the fixed trade deficit I evaluate real income as the money metric $\hat{I}_j = \hat{E}_j / \hat{P}_j$, where $E_j = w_j L_j + D_j$ is labor income plus the deficit, and (19) is its balanced-trade benchmark. The level change is $\Delta I = \sum_j I_j^0 (\hat{I}_j - 1)$.

Counterfactual tonnages The emissions, the revenue, and the damages all depend on the tonnage on the road, which is the shipped quantity rather than the delivered quantity. Under iceberg costs the truck carries τ_{oj}^k units to deliver one, so the value satisfies $X_{oj}^k = p_o^k (\tau_{oj}^k c_{oj}^k)$ and the shipped tonnage is X_{oj}^k / p_o^k . Its change is therefore the change in value deflated by the change in the domestic price, not the delivered price,

$$\hat{c}_{oj}^k = \frac{\hat{X}_{oj}^k}{\hat{p}_o^k}, \quad \hat{X}_{oj}^k = \hat{\lambda}_{oj}^k \hat{E}_j^k, \quad \hat{p}_o^k = \hat{w}_o \text{ (domestic)}, \quad \hat{p}_{mg}^k = 1 \text{ (import)}, \quad (20)$$

and the through-state tonnage follows from the fixed routing on each U.S. leg,

$$c_{rs}^{k'} = \hat{c}_r^k c_{rs}^k, \quad c_{rs}^{k'} = 0 \text{ unless } s \in \text{route}_r. \quad (21)$$

Deflating by the domestic price rather than by the delivered price $\hat{p}_o^k \hat{\tau}_{oj}^k$ is what makes (20) the road tonnage. The delivered (consumed) quantity falls with elasticity $-\sigma_k \beta_p^k$, while the shipped tonnage falls with elasticity $(1 - \sigma_k) \beta_p^k = b_k$, which is the same coefficient the gravity estimates deliver. Using the delivered quantity would overstate the reduction in emissions and revenue by the iceberg factor. I hold the routing fixed at the predetermined shortest-time path, so a state's emissions and revenue move only with the tonnage on the corridors that already cross it. The counterfactual tonnages $c_{rs}^{k'}$ feed (8), (10), (11), and (12), and the change in the welfare of a state is

$$\Delta W_s = \Delta I_s + \Delta T_s - \Delta \text{TSCC}_s - \Delta \text{LocalPollution}_s - \Delta \text{NonPollution}_s. \quad (22)$$

6 Estimation and Identification

I estimate a Poisson pseudo-maximum-likelihood gravity regression on the within-route movement of the route diesel price.

6.1 Specification

Taking (4) and (5) to the data, the estimating equation for commodity group k is, in multiplicative form,

$$X_{ij,t}^k = \exp\left(\delta_{i,t}^{X,k} + \delta_{j,t}^{M,k} + \psi_{ij}^k + b_k \ln \rho_{ij,t}\right) \nu_{ij,t}^k, \quad b_k \equiv (1 - \sigma_k) \beta_p^k. \quad (23)$$

I estimate it by PPML. PPML handles the zero flows that are common once trade is disaggregated to the commodity, origin, destination, and year. The estimation sample is the truck shipments (FAF mode 1). The highway diesel tax is imposed on trucking, the route price is built from the fuel cost, and the model's emissions and revenue accumulate to the truck leg, so I exclude the rail, water, and pipeline flows, which do not respond to the highway diesel price.⁵

The exporter-year effects $\delta_{i,t}^{X,k}$ and the importer-year effects $\delta_{j,t}^{M,k}$ are the outward and inward multilateral-resistance terms. They absorb every origin-specific and destination-specific shock in each year. The route effect ψ_{ij}^k absorbs every time-invariant bilateral determinant of trade, including distance, so b_k is identified from the within-route movement of the route diesel price over 2018 to 2024.

The coefficient b_k is the fuel-cost elasticity of trade. It is the product of the substitution elasticity and the diesel price coefficient, $(1 - \sigma_k) \beta_p^k$ which are not separately identified. The counterfactual flow and emission responses depend only on this product: the shipped-tonnage response is $\hat{\tau}^{1-\sigma_k} = \hat{\rho}_{ij}^{(1-\sigma_k)\beta_p^k} = \hat{\rho}_{ij}^{b_k}$. I calibrate σ_k from outside the price variation in Section 6.3 and recover $\beta_p^k = b_k / (1 - \sigma_k)$.

⁵Four commodities that move mostly by rail or pipeline (crude petroleum, coal, metallic ores, and other coal and petroleum products), each below 50% truck mode share with a clear gap to the next commodity at 51%, are dropped from the estimation. I keep them in the counterfactual, since their truck flows still burn fuel and pay tax.

6.2 Identification

To estimate β_p^k the route price must be unrelated to anything else that changes the trade flow on the same route. Three forms of endogeneity could break the estimate: reverse causality, measurement error, and omitted variables.

The first is that the freight itself moves the diesel price, as a route’s own trucks buy fuel. I test for it directly. Regressing the route price on the route’s own freight, net of route and year effects, gives -0.0003 (s.e. 0.0004): doubling a route’s volume shifts its price by about 0.02% . A single corridor is far too small a share of regional fuel demand to move its own price. It is consistent with the EIA decomposition, in which crude oil and taxes make up about two-thirds of the diesel price. The crude oil price is determined by global markets and the taxes are set by federal and state policy, so neither responds to one corridor’s demand.

The second is that carriers buy diesel where it is cheap rather than where they drive, so the constructed route price might not correspond to what they pay. The International Fuel Tax Agreement partially rules this concern out. A carrier owes each state tax in proportion to the miles it drives there, not the gallons it buys there, reconciled quarterly against the tax paid at the pump. Its tax cost is therefore the mileage-weighted sum of the state tax rates along the route, which is exactly the route price I construct, no matter where it fuels. Refueling moves the point of purchase but cannot avoid the mileage-apportioned tax. The only mismeasured part is the pre-tax wholesale difference, which varies little across nearby routes and is absorbed by the fixed effects.

The third is that an omitted factor moves both freight and the price: busy regions both ship more freight and burn more fuel. The origin-year and destination-year fixed effects absorb any shock common to an origin or a destination in a given year, and the route effect absorbs every time-invariant route characteristic, which removes this channel. What the fixed effects do not difference out is a shock specific to a single route in a single year, a bilateral confounder such as a corridor boom that raises the route’s freight and, through local fuel demand, its route price. Conditional on the fixed effects, the within-route movement in the

route price is driven principally by state diesel-tax changes, which legislatures set for fiscal reasons unrelated to any one corridor’s demand.

Identification rests on the within-route price movements that remain after the fixed effects, dominated by state tax changes, being uncorrelated with bilateral demand shocks. The concerns I have mentioned all push b_k toward zero: classical measurement error in the pre-tax price attenuates the coefficient, and any residual positive demand comovement pulls a negative coefficient up toward zero. The estimate is therefore conservative, a lower bound on the magnitude of the fuel-cost response. Because β_p^k scales every counterfactual trade and emissions response, this understates the policy effects. .

6.3 Trade elasticities and the recovered diesel price coefficients

I cannot separate σ_k from β_p^k within the price variation, so I take σ_k from outside the freight data. I use the sectoral trade elasticities of Caliendo and Parro (2015) and recover the diesel price coefficients as $\beta_p^k = b_k/(1 - \sigma_k)$.

6.4 Results

Table 4 reports the estimated elasticities. Seven of the ten groups have a correctly signed (negative) fuel-cost elasticity, and five of these are significant. The three wrong-signed groups, energy and fuels, chemicals, and machinery and hi-tech have high value density and I set them to zero. The largest recovered coefficients are in the fuel-intensive bulk groups: construction materials ($\beta_p = 0.08$), metals (0.06), and food (0.05). Agriculture and wood and are smaller (0.02 and 0.01). The pooled estimate is $\hat{b} = -0.045$ (s.e. 0.050) is small and imprecise as it reflects the level of aggregation from fuel-heavy bulk to value-dense manufactures.

The flow and emission responses are governed by the estimated b_k and do not depend on the choice of σ_k ; the welfare change scales with the recovered $\beta_p^k = b_k/(1 - \sigma_k)$. The recovered coefficients imply that a 20% increase in the route diesel price raises trade costs

Table 4: Fuel-cost elasticity of truck freight, by end-use group (separate PPML)

Group	$b_k = (1 - \sigma_k)\beta_p^k$	s.e.	σ_k	β_p^k	N
Construction/minerals	−0.193**	0.080	3.4	0.081	202,133
Metals	−0.321***	0.110	6.1	0.063	264,684
Food/beverage	−0.136***	0.034	3.6	0.052	303,863
Textiles/consumer	−0.276***	0.080	7.0	0.046	353,381
Agriculture	−0.165	0.105	10.1	0.018	177,004
Wood/paper	−0.180***	0.041	15.0	0.013	344,080
Mixed freight	−0.022	0.018	5.0	0.006	109,137
Energy/fuels	0.483	0.507	10	0	44,385
Chemicals/pharma	0.181*	0.103	4.1	0	405,762
Machinery/hi-tech	0.086	0.105	9.0	0	583,569

Separate PPML per group, exporter-year, importer-year, and route fixed effects, trade-type control; two-way clustered by route and year. $\beta_p^k = b_k/(1 - \sigma_k)$ with σ_k from Table B.1; wrong-signed high-value groups set to zero. Stars on b_k : * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

by at most about 1.5% (in construction).

7 Results

7.1 Increase in a federal tax

The federal diesel tax equals \$0.244 per gallon and has not changed since 1993. In this counterfactual I increase the federal tax up to its inflation-adjusted level in each year between 2018 and 2024 from \$0.18 per gallon in 2018 to \$0.29 in 2024. Table 5 reports the effect on real income, tax revenue, carbon and local-pollution damages, and the congestion, accident, and pavement externalities of trucking, and Table 6 reports the same quantities across states. I set the social cost of carbon to \$190 per ton.

Because every quantity outcome (diverted ton-miles, diesel burned, the carbon change, the local- and non-pollution co-benefits, and the tax revenue) depends only on the estimated fuel-cost elasticities b_k and not on the calibrated substitution elasticities σ_k , these figures are invariant to σ ; only the real-income measure responds, spanning −\$21 to −\$133 billion as we scale the calibrated σ vector from 0.5 to $2\times$ its baseline (Appendix C).

Table 5: Inflation adjustment of the federal tax, 2018–2024 (\$US mln)

Year	Δ tax (\$/gal)	Fiscal (transfers)		Environmental benefit		
		ΔI	Revenue	–SCC Δ CO ₂	– Δ LP	Non-poll.
2018	0.18	–5,534	6,226	37.6	13.5	39.5
2019	0.19	–5,718	6,384	39.7	13.5	41.7
2020	0.19	–5,711	6,330	45.5	14.6	47.8
2021	0.21	–6,395	7,090	39.1	13.7	41.1
2022	0.25	–7,537	8,357	30.9	10.9	32.4
2023	0.27	–8,134	9,110	45.4	16.5	47.7
2024	0.29	–8,629	9,714	60.2	21.3	63.2
Total		–47,657	53,211	298.4	104.0	313.6

Notes: Social cost of carbon is \$190. ΔI is the change in real income and Revenue is the tax revenue from diesel sales. $-\Delta$ LP is avoided local-pollution damage (PM_{2.5}, NO_x, VOC); Non-poll. is avoided congestion, accident, and pavement damage. The tax also falls on export legs, raising the delivered price of U.S. goods abroad; the corresponding foreign real-income change (Section 4.7) is –\$4.9 billion over 2018–2024.

The first result of this counterfactual is the decrease in carbon, local, and non-pollution externalities, shown in the last three columns of Table 5. Increasing the federal tax cuts carbon emissions by 1.57 million tons over the seven years, reduces local-pollutant damage by \$104 million, and reduces congestion, accident, and pavement damage by \$314 million. This result is driven by the reduction in total ton-miles: the higher diesel price raises the delivered cost of freight, and consumers respond by sourcing goods from closer origins. Figure 6 maps this reallocation on the highway network: the long-haul interior corridors lose freight as shipments shift toward shorter route. States buy fewer goods from remote areas, total fuel use falls, and every externality that scales with distance traveled falls with it.

The second result is that the non-pollution externalities are the largest source of environmental benefit. Congestion, accidents, and pavement damage account for \$314 million of avoided cost over the period, exceeding the \$298 million of avoided carbon damage and three times the reduction in local pollution. These costs are first-order in the freight-pricing literature (Austin (2015), Parry (2008)) but are absent from most trade-and-environment analyses; including them roughly doubles the measured benefit of the tax.

The third result comes from the trade model and explains why the tax raises more

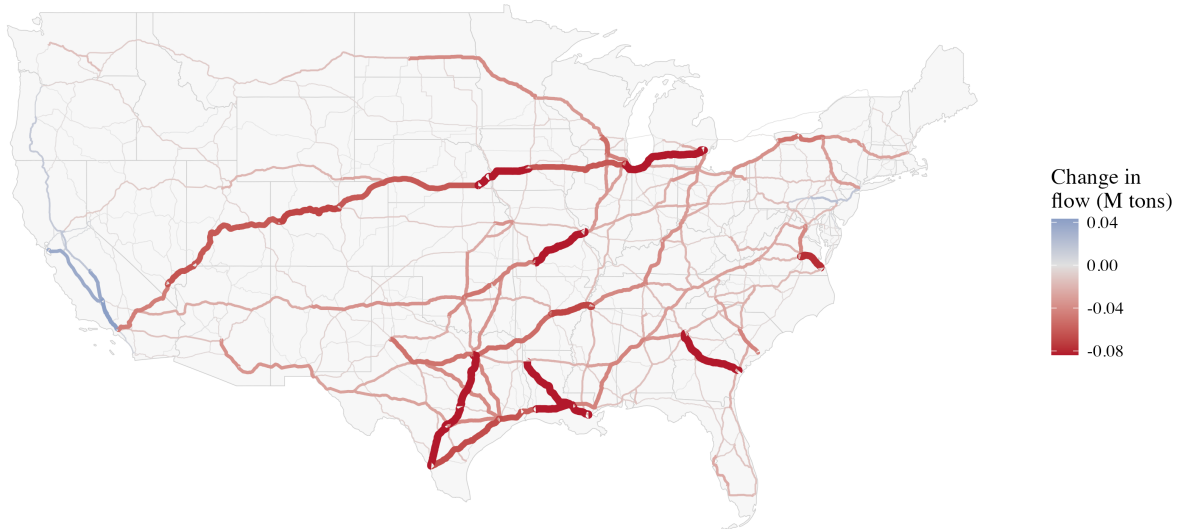


Figure 6: Change in freight flows from restoring the federal diesel tax (2019). Red corridors lose tonnage and blue corridors gain it; line width is proportional to the size of the change.

revenue than it costs US residents. Over the seven years the tax collects \$53.2 billion in revenue while real income falls by \$47.7 billion, so the revenue exceeds the domestic burden by \$5.5 billion. This gap is not paid by US residents. The tax is collected on every US truck leg, including the domestic leg of export shipments, and that leg raises the delivered price of US goods sold abroad. Because US export varieties are differentiated, foreign buyers have no perfect substitute and keep purchasing them, paying roughly \$5 billion of the tax rather than switching away.

The fourth result is the heterogeneity of these effects across space, reported in Table 6. The real-income loss is largest in the biggest consuming region: Texas (\$6.3 bln), California (\$4.5 bln), Illinois, and Florida. They buy the most freight and pay the most of the delivered-price increase. The environmental and fiscal gains, by contrast, concentrate in the freight-corridor states the shipments pass *through*. Texas, Louisiana, Illinois, Ohio, Tennessee, and

Missouri record the largest emission reductions and collect the most revenue, much of it on trucks that neither originate nor have their final destination there. Louisiana avoids \$22.5 million of carbon damage and collects \$1.35 billion in revenue while its residents lose less real income than those of states a fraction of its size; the emissions and revenue track the freight on the corridors that cross it, not the goods its residents buy. This is the transit externality the model is created to capture.

The reduction in local pollution is not universal. Figure 7 maps the change in local-pollution damage by county: most freight-corridor counties become cleaner, but the densest, highest-damage metropolitan counties become more polluted. California’s local-pollution damage *rises* by \$0.7 million even as the nation’s falls, because the tax reroutes some freight onto its already-congested corridors. Because the marginal damage of a pollutant depends on where it is emitted, a policy that reduces national emissions can still cause damage in densely populated states that absorb rerouted traffic.

7.2 Increase in a state tax: New Mexico

Figures B.1 and B.2 show that New Mexico has a large volume of through-traffic: freight that neither was produced in the state nor delivered to the state. This traffic imposes a pollution externality on New Mexico residents who live near the corridors even though these residents did not produce nor consume the goods shipped through their state. According to the research institute PSE Healthy Energy PSE Healthy Energy (2021), heavy- and medium-duty trucks made up 18 percent of the vehicle miles traveled in New Mexico in 2017, double the national average, and emitted 75 percent of the nitrogen oxides and fine particulate matter from on-road vehicles. A higher state tax might raise New Mexico’s welfare by curbing the pollution that comes from through-traffic by reducing ton-miles driven through the state.

In this counterfactual New Mexico raises its total state diesel tax from \$0.23 to California’s level of \$0.69 per gallon, an increase of \$0.46, applied in each year from 2018 to 2024.

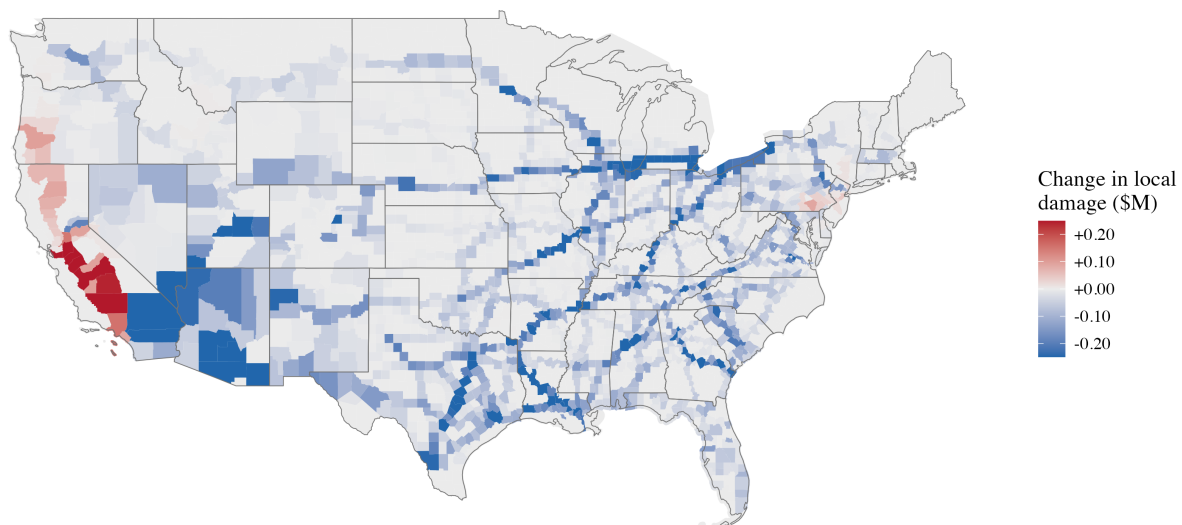


Figure 7: Change in local-pollution damage by county from the federal diesel tax, cumulative 2018–2024. Blue counties grow cleaner; red counties, the densest metropolitan areas, become more polluted as freight reroutes through them.

Table 7 reports the national and New Mexico effects, and Table 8 reports the effects across states.

The first result is that New Mexico reduces the externalities generated by transit shipment. Its own carbon emissions fall by the equivalent of \$24.9 million in damages over the period and its local pollution by \$5.2 million, while it collects \$2.15 billion in revenue, almost all of it on trucks that only pass through. The higher cost of crossing the state decreases the through-freight that generated the pollution, and the revenue is collected on the freight that continues to cross. Set against a real-income loss of \$317 million, New Mexico is a net winner.

The second result is that total pollution falls at the national level as well. Over the seven years national carbon emissions fall by \$68.3 million in damages, local pollution by \$20.5

million, and the congestion, accident, and pavement externalities by \$71.8 million. This is caused by freight demand responding to the higher delivered cost, so the reduction in shipments through the New Mexico corridor outweighs the rerouting, and the country as a whole emits less.

The third result is the spatial spillover in Table 8. The reduction concentrates on the New Mexico corridor: Arizona, Oklahoma, Texas, and California clean up along with New Mexico as the freight that used to cross the state falls away. Figure 8 shows this rerouting on the network: the Arizona–New Mexico corridor loses freight while the detour route to the north gain it. A set of Midwest and Plains states moves in the opposite direction. Iowa, Nebraska, Nevada, Minnesota, Kentucky, and Georgia each record an increase in carbon, local, and non-pollution damage as buyers substitute toward suppliers whose routes avoid New Mexico. Figure 9 maps the same reallocation in local-pollution damage: the corridor through Arizona, New Mexico, and southern California becomes cleaner while detour counties in the Plains and Midwest becomes more polluted. The increases are small next to the corridor’s reduction, so the national total falls, but they are real: a state acting alone to clean its own air pushes part of the burden onto the states it reroutes traffic through. This cross-border externality of a state-level fuel tax is invisible to any analysis that does not trace each shipment through the states it crosses.

8 Conclusion

This paper contributes to the ongoing discussion of optimal heavy-duty truck policies. I build a multi-commodity open-economy trade model on the US highway network that traces each shipment through the states and counties it crosses, and I use it to evaluate changes in the federal and state diesel taxes. I find significant heterogeneity in the effects of diesel regulation across space. When the federal tax is inflation-adjusted to its 1993 value, the real-income loss falls on the largest consuming states, while the environmental and fiscal

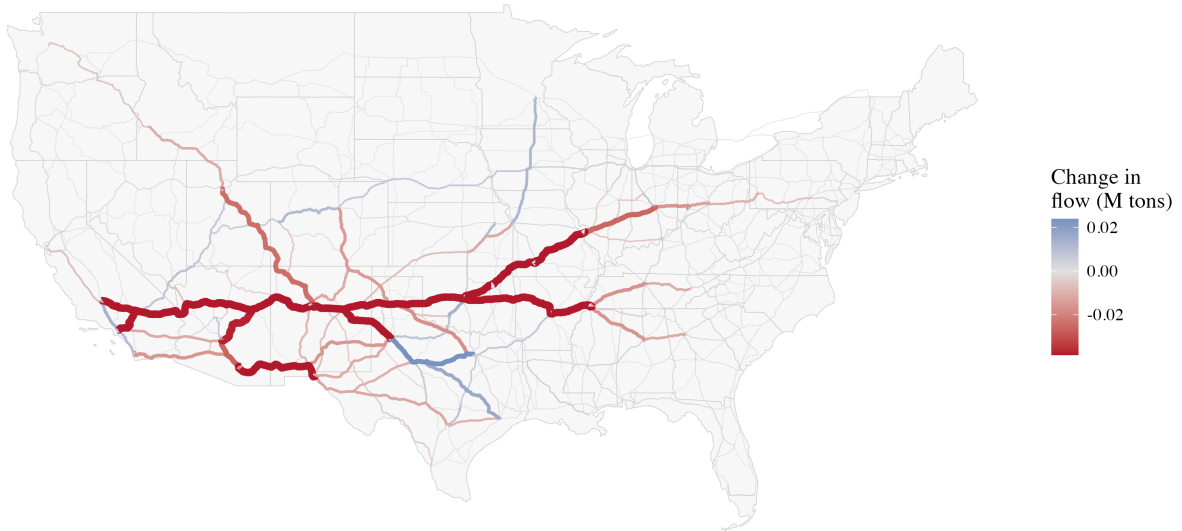


Figure 8: Change in freight flows from raising the New Mexico diesel tax to California's level (2019). The Arizona–New Mexico corridor (red) loses freight while the northern detour routes (blue) gain it.

gains accumulate to the corridor states that the freight passes through. This is the transit externality that the model is built to capture: a state experiences pollution, and collects revenue, from trucks that neither originate nor terminate within it.

Two further results follow from tracing the shipments rather than only their endpoints. First, the congestion, accident, and pavement externalities of trucking are the largest source of environmental benefit, larger than the avoided social cost of carbon and several times the reduction in local pollution. These costs are first-order in the freight-pricing literature but are usually left out of trade-and-environment analyses; including them roughly doubles the measured benefit of the tax. Second, because US export varieties are differentiated, part of the diesel tax is paid by foreign buyers on the domestic leg of export shipments, so the tax revenue exceeds the burden borne by US residents.

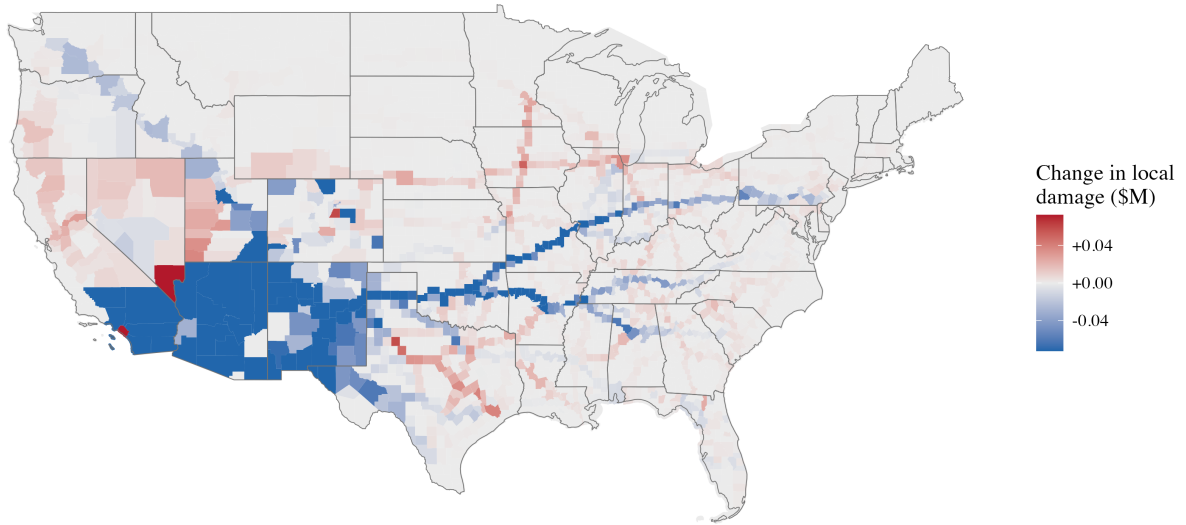


Figure 9: Change in local-pollution damage by county from the New Mexico tax increase, cumulative 2018–2024. The corridor through Arizona, New Mexico, and southern California grows cleaner (blue); detour counties in the Plains and Midwest become more polluted (red).

The state-level counterfactual shows why the federal and state margins cannot be analyzed in isolation. When New Mexico raises its tax to California’s level, it gains: the higher cost of crossing the state deters the through-freight that generated its pollution, and it collects revenue on the freight that continues to cross. National pollution falls as well, because freight demand responds to the higher delivered cost rather than simply rerouting onto longer paths. But the reduction is not shared evenly. The New Mexico corridor cleans up while a set of Plains and Midwest states absorbs the freight that reroutes around it and sees its own pollution rise. A state acting alone to clean its own air pushes part of the burden onto its neighbors. This cross-border externality is invisible to any analysis that does not follow each shipment through the states it crosses.

These conclusions depend on two features of the model. The physical route of each

shipment is held fixed at the predetermined shortest path, so the reallocation I measure is the change in where goods are sourced, not trucks choosing new highways; allowing price-responsive rerouting is a natural next step. The rest of the world is also fixed: foreign prices and foreign demand for US goods are unchanged and no foreign welfare is tracked, so the portion of the tax borne by foreign buyers is a statement about the US fiscal accounts rather than about global welfare.

This framework could be extended to analyze several other questions. What are the optimal state and federal diesel taxes once the transit externality and the non-pollution costs are priced? How does the equity-efficiency trade-off of diesel regulation look across states with very different freight exposure? And how do state policies on heavy-duty zero-emission vehicles affect welfare in the states their freight passes through? Answering these questions within the same spatial framework can provide new insight into the environmental regulation of heavy-duty trucks.

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Table 6: By-state effects of the federal diesel tax increase, 2018–2024 (\$US mln)

State	ΔI	Revenue	$-\text{SCC } \Delta \text{CO}_2$	$-\Delta \text{LP}$	$-\Delta \text{Non-Poll.}$
TX	−6306	7339	57.4	14.5	60.3
CA	−4454	3430	4.1	−0.7	4.3
IL	−2892	2978	12.8	6.1	13.4
FL	−2826	2366	6.0	2.2	6.3
CT	−2336	1001	2.3	0.0	2.5
PA	−2022	2101	6.0	2.2	6.3
OH	−1771	2307	11.2	5.7	11.7
GA	−1731	1637	10.6	5.3	11.2
NC	−1383	967	3.7	2.3	3.8
TN	−1214	1885	11.8	5.5	12.4
MI	−1179	807	6.5	3.4	6.9
MN	−1153	942	3.4	1.2	3.6
IN	−1054	1422	4.5	2.6	4.8
WI	−990	765	2.8	1.7	2.9
KY	−976	841	5.0	3.3	5.2
WA	−976	561	3.0	0.5	3.2
LA	−884	1350	22.5	7.9	23.7
AL	−824	1036	6.5	3.3	6.8
KS	−807	656	3.5	0.8	3.7
NY	−791	1143	5.9	1.5	6.2
UT	−788	916	6.6	1.7	6.9
CO	−786	1074	8.4	1.6	8.8
MD	−751	468	1.1	0.7	1.1
MA	−745	461	1.3	0.3	1.4
IA	−741	1287	5.8	2.4	6.1
AZ	−711	1069	6.4	2.5	6.7
OR	−692	723	1.2	0.1	1.3
OK	−624	1194	9.8	2.2	10.3
VA	−624	1405	8.8	4.7	9.2
SC	−604	706	4.1	2.4	4.3
AR	−482	912	7.6	3.2	7.9
NE	−476	1013	7.6	1.5	7.9
MS	−444	662	4.8	2.1	5.0
NV	−374	539	3.4	1.0	3.6
MO	−370	1540	10.5	3.7	11.0
ID	−344	425	1.6	0.3	1.7
ND	−329	326	2.6	0.5	2.7
MT	−232	418	3.0	0.4	3.2
NM	−221	972	7.6	1.7	8.0
WV	−211	444	1.7	0.8	1.8
SD	−181	226	1.2	0.3	1.3
ME	−120	61	0.1	0.0	0.1
WY	−115	563	3.1	0.5	3.3
VT	−65	46	0.2	0.0	0.2
DE	−33	43	0.1	0.0	0.1
NH	−27	19	0.1	0.0	0.1
Total	−47657	53044	298.0	103.7	313.2

Notes: Social cost of carbon is \$190. Revenue is the diesel-tax take collected in the state, including tax on through-traffic. A negative $-\Delta \text{LP}$ entry (California) is an *increase* in local pollution. Hawaii and Alaska excluded; Rhode Island and New Jersey merged into neighbors.

Table 7: Raising the New Mexico diesel tax to California’s level, 2018–2024 (\$US mln)

Year	Δ tax (\$/gal)	Fiscal		Environmental benefit		
		ΔI	Revenue	–SCC Δ CO ₂	– Δ LP	– Δ Non-Poll.
2018	0.40	–228	253	8.8	2.6	9.2
2019	0.46	–259	287	10.4	3.1	10.9
2020	0.47	–256	281	12.5	3.7	13.2
2021	0.44	–242	264	9.0	2.7	9.4
2022	0.45	–253	278	6.3	1.9	6.6
2023	0.53	–295	328	8.3	2.5	8.7
2024	0.73	–410	454	13.0	3.9	13.7
Total (US)		–1941	2146	68.3	20.5	71.8
<i>of which New Mexico</i>		–317	2146	24.9	5.2	26.1

Notes: Social cost of carbon is \$190. New Mexico raises its total state tax from \$0.23 to California’s level each year; the increment rises from \$0.40 in 2018 to \$0.73 in 2024 as California’s tax climbs. Revenue is collected entirely by New Mexico, on through-traffic. The tax also raises the delivered price of U.S. exports whose routes cross New Mexico; the implied foreign real-income change is –\$188 million over 2018–2024. The national totals are net of the spillover in Table 8.

Table 8: Spatial spillover of the New Mexico tax increase, cumulative 2018–2024 (\$US mln)

State	ΔI	–SCC Δ CO ₂	– Δ LP	– Δ Non-Poll.
<i>New Mexico corridor: pollution falls</i>				
New Mexico	–317	24.9	5.2	26.1
Arizona	–177	14.5	4.7	15.2
Oklahoma	–31	9.0	1.6	9.5
Texas	–239	8.7	1.2	9.2
California	–479	5.3	4.8	5.5
Missouri	–27	2.7	1.0	2.8
<i>Detour corridors: pollution rises</i>				
Iowa	0.6	–1.0	–0.4	–1.0
Nebraska	–0.4	–0.7	–0.1	–0.8
Nevada	–21	–0.4	–0.1	–0.5
Minnesota	–2	–0.4	–0.2	–0.4
Kentucky	–37	–0.3	–0.2	–0.3
Georgia	–52	–0.3	–0.1	–0.3

Notes: Social cost of carbon is \$190. The six states with the largest pollution decrease and the six with the largest increase. A negative (purple) entry is added damage as freight reroutes off the New Mexico corridor. The remaining 34 states have effects below \$0.28 mln.

A Appendix

B Derivation of the commodity-level gravity equation

I solve the consumer's problem in two stages. The upper stage splits total expenditure E_j across the K commodities; the lower stage splits each commodity's expenditure across origins. Throughout, the delivered price of commodity k from origin o is $p_{oj}^k = \tau_{oj}^k p_o^k$.

Stage 1: allocation across commodities Let P_j^k denote the price index of the commodity- k composite C_j^k (derived in Stage 2 below). The consumer chooses the composites to maximize the Cobb–Douglas aggregator subject to the budget,

$$\max_{\{C_j^k\}} \prod_{k=1}^K (C_j^k)^{\alpha_j^k} \quad \text{s.t.} \quad \sum_{k=1}^K P_j^k C_j^k = E_j. \quad (24)$$

The first-order condition for C_j^k is $\alpha_j^k U_j / C_j^k = \mu P_j^k$, where μ is the multiplier on the budget, so $P_j^k C_j^k = \alpha_j^k U_j / \mu$. Summing over k and using $\sum_k \alpha_j^k = 1$ gives $\sum_k P_j^k C_j^k = U_j / \mu = E_j$, hence $U_j / \mu = E_j$ and

$$E_j^k \equiv P_j^k C_j^k = \alpha_j^k E_j. \quad (25)$$

Cobb–Douglas therefore assigns a constant share α_j^k of expenditure to each commodity.

Stage 2: allocation across origins Given the expenditure E_j^k on commodity k , the consumer attains the composite C_j^k at least cost. Equivalently I minimize the expenditure needed to deliver one unit of C_j^k ,

$$\min_{\{c_{oj}^k\}} \sum_{o \in \mathcal{O}} p_{oj}^k c_{oj}^k \quad \text{s.t.} \quad \left[\sum_{o \in \mathcal{O}} (c_{oj}^k)^{\frac{\sigma_k - 1}{\sigma_k}} \right]^{\frac{\sigma_k}{\sigma_k - 1}} = C_j^k. \quad (26)$$

The first-order condition for c_{oj}^k , with multiplier λ on the constraint, is

$$p_{oj}^k = \lambda (C_j^k)^{\frac{1}{\sigma_k}} (c_{oj}^k)^{-\frac{1}{\sigma_k}}, \quad (27)$$

where I used $[\sum_o (c_{oj}^k)^{\frac{\sigma_k-1}{\sigma_k}}] = (C_j^k)^{\frac{\sigma_k-1}{\sigma_k}}$. Solving for the demand,

$$c_{oj}^k = C_j^k \lambda^{\sigma_k} (p_{oj}^k)^{-\sigma_k}. \quad (28)$$

Substituting (28) into the expenditure on commodity k ,

$$E_j^k = \sum_{o \in \mathcal{O}} p_{oj}^k c_{oj}^k = C_j^k \lambda^{\sigma_k} \sum_{o \in \mathcal{O}} (p_{oj}^k)^{1-\sigma_k}. \quad (29)$$

Define the commodity- k price index as the unit cost of the composite,

$$P_j^k = \left[\sum_{o \in \mathcal{O}} (p_{oj}^k)^{1-\sigma_k} \right]^{\frac{1}{1-\sigma_k}}. \quad (30)$$

Because the minimized expenditure equals $P_j^k C_j^k$ by definition, comparing the last two lines gives $\lambda = P_j^k$: the multiplier is the marginal cost of the composite, which is exactly the price index. The demand (28) then becomes

$$c_{oj}^k = C_j^k \left(\frac{p_{oj}^k}{P_j^k} \right)^{-\sigma_k}. \quad (31)$$

Trade value. The value of area j 's purchases of commodity k from origin o is, using (31) and $C_j^k P_j^k = E_j^k$,

$$X_{oj}^k \equiv p_{oj}^k c_{oj}^k = C_j^k P_j^k \left(\frac{p_{oj}^k}{P_j^k} \right)^{1-\sigma_k} = \left(\frac{p_{oj}^k}{P_j^k} \right)^{1-\sigma_k} E_j^k. \quad (32)$$

Substituting the delivered price $p_{oj}^k = \tau_{oj}^k p_o^k$, the index (30), and the commodity expenditure (25) $E_j^k = \alpha_j^k E_j$ yields the commodity-level gravity equation (4),

$$X_{oj}^k = \left(\frac{\tau_{oj}^k p_o^k}{P_j^k} \right)^{1-\sigma_k} E_j^k, \quad P_j^k = \left[\sum_{o \in \mathcal{O}} (\tau_{oj}^k p_o^k)^{1-\sigma_k} \right]^{\frac{1}{1-\sigma_k}}. \quad \blacksquare \quad (33)$$

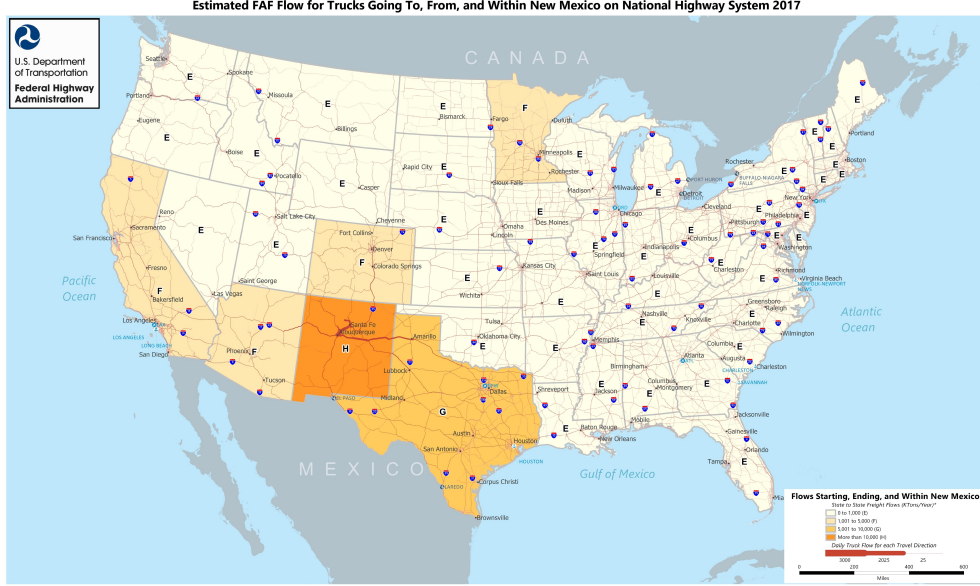


Figure B.1: Trucking to/from New Mexico (*Source: FAF5.5*)

Table B.1: Commodity-group trade elasticities (Caliendo and Parro (2015))

Group	Caliendo–Parro sector(s)	θ_k	$\sigma_k = \theta_k + 1$
Agriculture	Agriculture	9.1	10.1
Food/beverage	Food	2.6	3.6
Construction/minerals	Minerals (ores: Mining)	2.4	3.4
Energy/fuels	Petroleum	64.9	65.9
Chemicals/pharma	Chemicals, Plastic, Medical	3.1	4.1
Wood/paper	Wood, Paper	14.0	15.0
Metals	Basic metals, Metal products	5.1	6.1
Textiles/consumer	Textile, Other	6.0	7.0
Machinery/hi-tech	Machinery, Electrical, Auto	8.0	9.0
Mixed freight	Other	4.0	5.0

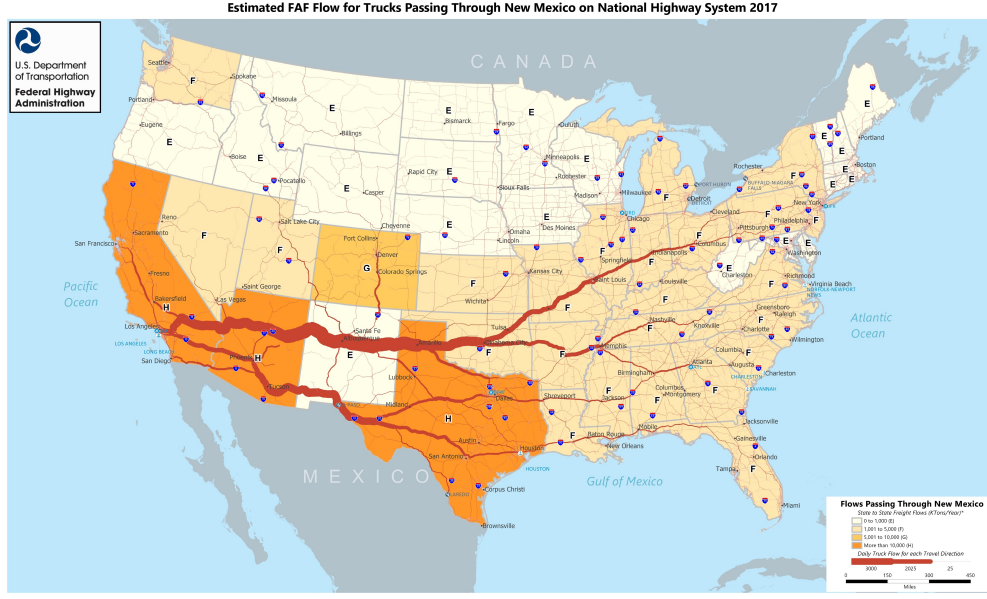


Figure B.2: Trucking through New Mexico (*Source: FAF5.5*)

C Robustness checks

The counterfactual results depend on two elasticities. The fuel-cost elasticities b_k of Table 4 are *estimated* and drive the physical response of freight flows; the substitution elasticities σ_k are *calibrated* from Caliendo and Parro (2015) and enter only through the price index. Every quantity outcome (diverted ton-miles, diesel burned, the carbon change, the local-pollution and non-pollution co-benefits, and the tax revenue) is a function of b_k alone and is therefore invariant to σ . Only the real-income (welfare) measure depends on σ , and it does so through two channels that both carry $1 - \sigma$ in a denominator: the implied cost pass-through $\beta_p^k = b_k/(1 - \sigma_k)$ and the welfare exponent $\alpha/(1 - \sigma_k)$ in $\hat{I}_j = \prod_k (\hat{\lambda}_{jj}^k)^{\alpha/(1 - \sigma_k)}$.

To check the sensitivity of the welfare figure, I scale the entire calibrated σ vector by a common factor between 0.5 and 2, holding the estimated b_k fixed and preserving the cross-commodity ranking of Table 4 (construction and food remain the most substitutable, wood and paper the least). Table C.3 reports the resulting real-income change.

The pattern is the standard gravity result that welfare effects scale with $1/(\sigma - 1)$: the smaller the substitution elasticity, the larger the estimated loss. Low σ means goods

are poor substitutes, so a fuel-cost shock cannot be routed around cheaply and its welfare incidence is large; high σ lets shippers substitute across origins at little cost, and the welfare effect collapses. Across the range considered the real-income change spans $-\$133$ billion to $-\$21$ billion.

Table B.2: Effect of the New Mexico tax increase on all states, 2018–2024 (\$US mln)

State	ΔI	$-\text{SCC } \Delta \text{CO}_2$	$-\Delta \text{LP}$	Non-poll.
NM	−316.6	24.87	5.15	26.14
AZ	−177.5	14.46	4.69	15.20
OK	−31.1	9.01	1.56	9.47
TX	−239.4	8.72	1.22	9.17
CA	−479.5	5.28	4.79	5.55
MO	−26.9	2.69	0.96	2.83
AR	−15.5	2.35	0.97	2.47
TN	−62.5	1.21	0.52	1.27
CO	−33.7	1.05	0.35	1.10
UT	−150.6	0.74	0.25	0.78
ID	−5.5	0.61	0.11	0.64
IN	−41.6	0.52	0.28	0.55
AL	−16.7	0.37	0.20	0.39
IL	−26.8	0.37	0.39	0.38
MS	−5.4	0.29	0.16	0.30
WA	−22.6	0.20	0.06	0.21
PA	−37.0	0.19	0.20	0.20
OR	−6.5	0.16	0.02	0.17
OH	−14.3	0.09	0.12	0.10
ME	0.2	0.00	0.00	0.00
NH	−0.1	0.00	0.00	0.00
VT	0.3	0.00	0.00	0.00
DE	−0.3	−0.01	−0.01	−0.01
MA	−0.2	−0.02	−0.01	−0.02
MI	−6.8	−0.02	0.01	−0.02
ND	0.1	−0.03	−0.01	−0.03
CT	−8.1	−0.03	0.00	−0.03
SD	−0.2	−0.05	−0.01	−0.05
VA	−7.8	−0.05	−0.01	−0.06
WV	−0.6	−0.06	−0.02	−0.06
KS	−5.1	−0.06	−0.01	−0.07
MT	−0.1	−0.07	−0.01	−0.08
MD	−20.5	−0.08	−0.03	−0.08
NY	3.0	−0.09	−0.02	−0.10
WI	−2.4	−0.12	−0.05	−0.13
NC	−19.2	−0.18	−0.09	−0.19
SC	−12.0	−0.18	−0.10	−0.19
LA	−12.6	−0.22	−0.07	−0.23
FL	−24.3	−0.25	0.01	−0.26
WY	−2.5	−0.26	−0.04	−0.28
GA	−52.4	−0.28	−0.08	−0.29
KY	−37.2	−0.30	−0.16	−0.31
MN	−2.3	−0.36	−0.16	−0.38
NV	−20.9	−0.44	−0.12	−0.46
NE	−0.4	−0.73	−0.14	−0.77
IA	0.6	−0.97	−0.35	−1.02
Total (US)	−1941	68.3	20.5	71.8

Notes: Social cost of carbon is \$190. A positive entry is avoided damage (pollution falls); a negative entry is added damage as freight reroutes off the New Mexico corridor onto that state. Revenue is collected entirely by New Mexico (\$2,146 mln) and is omitted. Hawaii and Alaska excluded; Rhode Island and New Jersey merged into neighbors.

Table C.3: Sensitivity of welfare to the substitution elasticity (\$US bn)

σ scenario	ΔI	Revenue	Env. benefit	Total W
$\sigma_k \times 0.5$	-133.0	53.2	0.7	-79.1
$\sigma_k \times 0.75$	-69.8	53.2	0.7	-15.9
σ_k (Table 4)	-47.7	53.2	0.7	+6.2
$\sigma_k \times 1.5$	-29.2	53.2	0.7	+24.7
$\sigma_k \times 2.0$	-21.1	53.2	0.7	+32.8

Notes: Cumulative 2018–2024, federal counterfactual. ΔI is the change in real income from (19) scaled with the substitution elasticities σ_k of Table 4. Revenue and the environmental benefit (avoided carbon, local, and non-pollution damages, \$0.3 + \$0.1 + \$0.3 bn) depend only on the estimated b_k and are invariant to σ . Total welfare is $W = \Delta I + \text{Revenue} + \text{Env. benefit}$. At low σ_k this priced loss exceeds the revenue and total welfare is negative.

Table C.4: Sensitivity of welfare to the substitution elasticity, New Mexico counterfactual (\$US mln)

σ scenario	ΔI	Revenue	Env. benefit	Total W
$\sigma_k \times 0.5$	-5405	2146	160.6	-3098
$\sigma_k \times 0.75$	-2843	2146	160.6	-536
σ_k (Table 4)	-1941	2146	160.6	+366
$\sigma_k \times 1.5$	-1191	2146	160.6	+1116
$\sigma_k \times 2.0$	-860	2146	160.6	+1447

Notes: Cumulative 2018–2024, New Mexico raising its diesel tax to California’s level. ΔI is the change in *national* real income from (19) scaled with the substitution elasticities σ_k of Table 4. Revenue and the environmental benefit (avoided carbon, local, and non-pollution damages) depend only on the estimated b_k and are invariant to σ . Total welfare is $W = \Delta I + \text{Revenue} + \text{Env. benefit}$.